VAIBHAV KANDPAL

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EDUCATION

IIT Roorkee Feb 2026 – Present
Professional Certificate in Data Science & AI (6-month intensive) *Remote*

Dr. Akhilesh Das Gupta Institute of Professional Studies (GGSIPU) Aug 2024 – May 2028 (Expected)
Bachelor of Technology in Computer Science and Engineering *New Delhi, India*

AWARDS & ACHIEVEMENTS

- NASA Space Apps Challenge 2024: Top 5 in India (823 competing teams); built OrbitAir real-time air quality forecasting system integrating TEMPO satellite, EPA, OpenAQ & NOAA feeds - github.com/Vaibhav20k/ORBIT-AIR
- Paytm agent{a}thon winner - first place
- Deloitte Australia Data Analytics Job Simulation, Forage certified (Jul 2025)

EXPERIENCE

Cimplifie | Software Developer Intern Jun 2026 – Jul 2026 (In-office)

- Shipped 4 client-facing React dashboards consuming internal REST APIs, reducing average client reporting turnaround from 3 days to same-day by replacing manual Excel exports with live data views.
- Integrated GitHub Actions CI/CD pipeline for the company's Node.js backend automating lint, test, and Vercel preview deploy on every PR, cutting QA review time by 50%.
- Wrote browser-automation scripts in Python to validate UI flows across staging environments after each release, catching 11 regression bugs before they reached production.

PROJECTS

OrbitAir: AI-Based Air Quality Forecasting System | *Python, React, TimescaleDB, FastAPI, Docker*
github.com/Vaibhav20k/ORBIT-AIR

- Built a multi-source ETL pipeline fusing TEMPO satellite, EPA, OpenAQ & NOAA feeds into TimescaleDB via async batch processing, reducing pipeline runtime from 8 min to under 2 min (4× throughput gain).
- Built a Random Forest forecasting model with temporal train-test validation and SHAP-based explainability to surface the top 5 regional pollution drivers, visualized on a React + Leaflet dashboard.
- Refactored React component tree with lazy-loaded map tiles; improved Lighthouse performance score from 61 → 89. Built and deployed end-to-end in 48 hours for NASA Space Apps Challenge.

DualSentry: Fraud & Anomaly Detection System | *Go, Python, FastAPI, PostgreSQL, Apache Kafka, Docker, XGBoost, gRPC, Redis*
github.com/Vaibhav20k/DualSentry-model

- Built an event-driven fraud detection platform using Go microservices, gRPC, Apache Kafka, PostgreSQL, and Docker, processing transactions through a real-time pipeline with feature engineering, ML inference, and fraud decisioning (ALLOW / REVIEW / BLOCK) while persisting predictions for downstream analytics.
- Built an ML anomaly detection engine using XGBoost and Isolation Forest with 20+ engineered behavioral features including transaction velocity, amount deviation, merchant frequency, device history, location risk, and active-hour analysis; implemented automated CSV dataset generation for continuous model retraining and evaluation.
- Developed an end-to-end fraud intelligence workflow integrating feature extraction, baseline profiling, model inference, decision engine, and prediction persistence, achieving 100% successful end-to-end transaction processing from gRPC ingestion through Kafka streaming to PostgreSQL-backed fraud prediction storage with versioned model metadata.

BehaviourIQ: Local-First Behavioral Intelligence Platform | *Rust, Python, PyTorch Geometric, FAISS, LangGraph, SQLite*
github.com/Vaibhav20k/BehaviourIQ

- Engineered a local-first behavioral intelligence engine in Rust with a WAL-mode SQLite behavior graph, ingesting Linux auth, GitHub, and Docker event logs through 4 extensible adapters into a unified Entity-Event-Context-Time graph model.
- Trained GraphSAGE Graph Neural Networks (PyTorch Geometric) over multi-layer temporal baselines (24h-365d horizons) with Jensen-Shannon divergence drift metrics, fused with statistical and velocity signals for explainable anomaly detection with zero cloud dependency.
- Built a LangGraph-based investigation agent with FAISS semantic retrieval over incident embeddings, enabling evidence-grounded natural language querying and automated Markdown/PDF incident reporting with no hallucinated findings.

TECHNICAL SKILLS

Languages: Python, JavaScript, SQL, C, C++, Rust
Frontend: React.js, Next.js, Tailwind CSS
Backend: FastAPI, Node.js, Django, REST APIs, WebSockets
Databases: PostgreSQL, MySQL, SQLite, TimescaleDB, Redis
ML & AI: PyTorch Geometric, TensorFlow, Graph Neural Networks, LangGraph/LangChain, Scikit-learn, XGBoost, FAISS (RAG), SHAP, Feature Engineering
DevOps & Tools: Docker, Git, GitHub Actions CI/CD, Microservices, API Design
Data Science: ETL, Data Pipeline, Distributed Systems, Data Warehouse